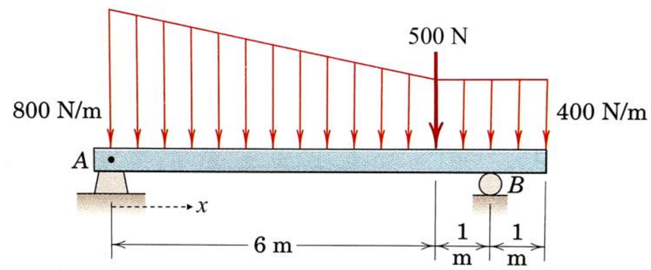
Section 2 – Shear and Moment Diagrams

Provide information for constructing the ***shear diagram***. The support reactions are A_y = 2300 N acting up, B_y = 2600 N acting up. A is a pin and B is a roller.

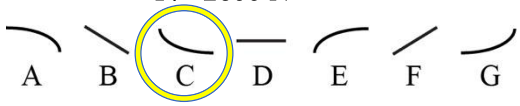


1. (3 pt) What happens to the shear diagram at *x* = 6 m?
   A. Drop 500 N
   B. Jump 2300 N
   C. Jump 2600 N
   D. Drop 400 N
   E. Jump 500 N
   F. Drop 800 N

What is the numerical value (magnitude only) of the change in shear for the region 0 < *x* < 6 m and what is the behavior (shape) of the shear diagram for the region 0 < *x* < 6 m?

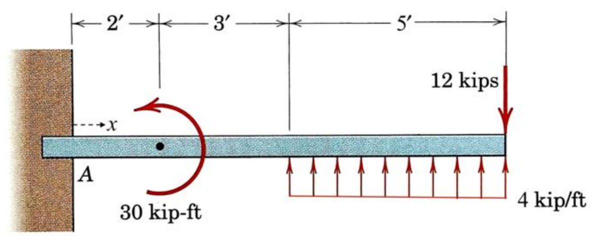
2. (3 pt) Change = __________ (magnitude only)
   A. 3600 N
   B. 2300 N
   C. 1200 N
   D. 2400 N
   E. 500 N
   F. 2600 N

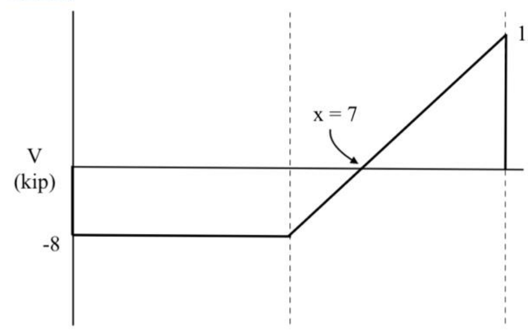
3. (3 pt) Behavior:



4. (3 pt) Determine the order of the equation for the internal shear for the region 6 < *x* < 7 m.
   A. first order, linear
   B. zero
   C. constant
   D. second order, parabolic
   E. third order cubic

Provide information for constructing the ***moment diagram***. The support reactions are A_y = 8 kip acting down, M_A = 60 kip-ft acting clockwise. A is a fixed support. The shear diagram is provided.

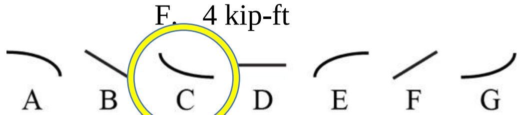




What is the numerical value (magnitude only) of the change in moment for the region 5 < *x* < 7 ft and what is the behavior (shape) of the moment for the region 5 < *x* < 7 ft?

5. (3 pt) Change = __________ (magnitude only)
   A. 8 kip-ft
   B. 5 kip-ft
   C. 20 kip-ft
   D. 12 kip-ft
   E. 30 kip-ft
   F. 4 kip-ft

6. (3 pt) Behavior:



7. (3 pt) What happens to the moment diagram at *x* = 2 ft?
   A. Drop 30 kip-ft
   B. Drop 16 kip-ft
   C. Jump 30 kip-ft
   D. Drop 8 kip-ft
   E. Jump 16 kip-ft
   F. Drop 60 kip-ft

8. (4 pt) Determine the equation for the internal bending moment for the region 2 < *x* < 5 ft.
   A. -8x + 30
   B. -8x + 60
   C. $8x^2 - 16x$
   D. $8x^2 + 24x$
   E. 40x